

Hemanth Chandravamsi

Postdoctoral fellow, Johns Hopkins University
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Research Interests

Numerical analysis · Computational fluid dynamics (CFD) · High-speed flows · Data assimilation · Supersonic jet noise · High performance computing (HPC) · Shock boundary layer interaction · Physics informed machine learning; PINNs

Education

Ph.D. Mechanical Engineering 2021-2024
Technion - Israel Institute of Technology Haifa, Israel
Thesis: *Resolution optimized GPU-accelerated numerical schemes for high-speed flows: application to supersonic jet noise*
Advisor: Prof. Steven H. Frankel

M.Tech Aerospace Engineering 2018-2020
Indian Institute of Technology Madras Chennai, India
Thesis: *Control of protrusion induced unsteady three-dimensional shock boundary layer interaction using suction*
Advisor: Prof. R. Sriram

B.Eng. (Hons.) Manufacturing Engineering (Minor in Material Science) 2013-2017
Birla Institute of Technology and Science, Pilani Hyderabad, India

Appointments

Postdoctoral fellow Jan 2026 – Present
Johns Hopkins University Baltimore, MD, USA
Data assimilation for high-speed flows and latent-space nonlinear dynamics; differentiable flow solvers, adjoint sensitivity, and direct simulation Monte Carlo (DSMC).
Host: Tamer A. Zaki, Flow Science & Engineering Lab

Postdoctoral fellow Oct 2024 – Jan 2026
Technion - Israel Institute of Technology Haifa, Israel
Numerical characterization of unstart in scramjet intake using high-fidelity large eddy simulations.
Host: Steven H. Frankel, CFDLAB Technion

Project Associate Oct 2020 – Feb 2021
Indian Institute of Technology Madras Chennai, India
Unsteady dynamics of three-dimensional shock boundary layer interaction induced separation
Host: R. Sriram, Boundary layer and separation dynamics lab.

Scholarships/Awards

1. **Seal of Excellence**, Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship, European Commission (2026). Awarded to proposals scoring above all evaluation thresholds; evaluation score 91/100.
2. Tuition waiver and full graduate scholarship for Ph.D. studies, Technion, 2021-2024.
3. **Half-Time Research Assistantship (HTRA)**, Ministry of Education (formerly MHRD), Government of India, for M.Tech. studies at IIT Madras, 2018-2020.

Research Experience

Johns Hopkins University, Department of Mechanical Engineering

As postdoctoral fellow

Baltimore, MD, USA

Advisor: Prof. Tamer A. Zaki

● **Data Assimilation for High-Speed Flows** (Ongoing): Development of a feature-preserving latent-space ensemble Kalman filter (Latent-EnKF) for assimilating sparse observations into shock-containing flows, preserving discontinuities that conventional latent representations smear.

● **Differentiable Solvers and Adjoint Sensitivity** (Ongoing): Adjoint-based sensitivity analysis for compressible flow solvers, and direct simulation Monte Carlo (DSMC) for rarefied and transitional-regime flows.

Technion, Faculty of Mechanical Engineering

As postdoctoral fellow

Haifa, Israel

Advisor: Prof. Steven H. Frankel

● **Scramjet Un-start Phenomenon**: Development of in-house tools for dynamic mesh simulation to study the un-start phenomenon prevalent in scramjet engines using GPU-accelerated large eddy simulations.

● **Implicit Neural Representation Networks**: Proposed SOS-SIREN, a *noisy sinusoidal representation neural network* for reconstructing complex audio and image signals with dominant high-frequency features as continuous neural functions. Applications include compressing large fluid flow datasets and solving forward problems via Physics Informed Neural Networks.

● **Method and Solver Development**: (a) Development of a neural network based optimal order detection model for high-resolution compressible flow simulations using polynomials of different orders and spectral resolutions. (b) Development of a new unstructured flux reconstruction (nodal Discontinuous Galerkin) based compressible Navier-Stokes solver with artificial diffusion based shock-capturing capability.

● **Shock Wave Dynamics**: Collaborative work with Prof. Omri Ram and the TFML laboratory at Technion on investigating the dynamics of shock waves passing through roughness elements in confined tubes.

As graduate student (Ph.D.)

Advisor: Prof. Steven H. Frankel

● **Method Development**: (a) Proposed a new resolution-optimized high-order *dispersion-relation-preserving* scheme for non-linear diffusion terms; demonstrated state-of-the-art accuracy and improved stability. (b) Investigated the use of spatially *adaptive modal filtering* for resolving discontinuities in compressible flows through the flux reconstruction framework.

● **Solver Development (from scratch)**: (a) **MIGHTY**: Built a multi-GPU accelerated, curvilinear multi-block Navier-Stokes solver with state-of-the-art shock-capturing and diffusion schemes. (b) **CAF-FR**: Developed from scratch a Flux reconstruction solver for Euler equations with shock capturing capability. (c) **MIGHTY-KEEP**: Co-developed a kinetic energy and entropy-preserving compressible Navier-Stokes solver. (d) **LEGOS**: Wrote a Fortran-based grid generator for simple multi-block geometries.

● **Physics Informed Machine Learning**: One of the lead developers of PINNACLE - a GPU-accelerated physics informed neural network (PINNs) framework to solve forward and inverse problems relevant to Engineering.

● **Supersonic Jet Noise**: Conducted Large Eddy Simulations to study the influence of jet temperature on shock-cell spacing and shear-layer convection velocity at a Reynolds number $O(10^5)$. Performed modal analysis of screech dynamics across varying temperature ratios. Investigated the influence of impingement wall curvature on flow unsteadiness and screech characteristics for convex and concave impinging wall geometries.

Indian Institute of Technology Madras

As graduate student (M.Tech.)

Chennai, India

Advisors: R. Sriram

● **3D Shock-Boundary Layer Interaction and Control**: Conducted wind tunnel experiments to analyze separation length scaling laws in protuberance-induced shock-boundary layer interactions across various geometries. Successfully demonstrated separation and unsteadiness reduction through *boundary layer bleed* via suction.

● **Experimental Rigs**: Designed a Mach 3 supersonic wind tunnel to study *protuberance-induced 3D shock-boundary layer interaction*. Integrated a suction chamber for passive flow control. Developed a vernier caliper-assisted precision Pitot tube for boundary layer characterization. Arranged flow visualization setup.

External Collaborations

1. Cambridge University & Indian Institute of Technology Madras

Collaborators: Nagabhushana R. Vadlamani and Paul G. Tucker

● **High-Order Eikonal Solver**: Developed a high-resolution Eikonal and Hamilton-Jacobi equation solver for computing *wall distances* using high-order Padé schemes and spatially adaptive artificial viscosity.

2. Transient Fluid Mechanics Laboratory, Technion – Israel Institute of Technology

Collaborators: Omri Ram, Jibu Tom Jose, and Yoav Gichon (experimentalists)

- **Shock Wave Dynamics:** Developed a Lagrangian *Geometric Shock Dynamics* framework for computing spatially evolving shock fronts. Conducted *Large Eddy Simulations* to characterize temporal dynamics of shock waves passing through an abrupt area change and derived an empirical scaling law for estimating steady-state expanded shock strength.

Publications & Open-Source Repositories

● Compressible flows · ● Numerical analysis · ● Machine Learning · ● Software/HPC

Google Scholar (September 2026): 127 citations, h-index 8, i10-index 6.

Journal Articles & Preprints

[†]Equally contributed authors

- R. Heppner, **H. Chandravamsi**, Y. Gichon, S. H. Frankel, & O. Ram (2026). *Shock propagation through a local constriction*. Journal of Fluid Mechanics (*accepted for publication*). [PDF]
- **H. Chandravamsi**, H. Hu, P. Thiagarajan, & T. A. Zaki (2026). *Feature-preserving Latent-EnKF for data assimilation of flows with shocks*. Preprint under review. [PDF]
- **H. Chandravamsi**, D. V. Shenoy, & S. H. Frankel (2026). *Supersonic jet impingement on concave surfaces*. Preprint under review. [PDF]
- Z. Chen[†], **H. Chandravamsi**[†], S. Pisnoy[†], A. Goldgewert, G. Shaviner, B. Shrager, & S. H. Frankel (2026). *Hybrid quantum-classical PINNs for scientific computing: a multi-GPU open-source framework*. arXiv preprint. [PDF]
- **H. Chandravamsi**, S. H. Frankel (2024). *High resolution optimized high-order schemes for discretization of non-linear straight and mixed second derivative terms*. Journal of Computational Physics, 513, 113170. [PDF]
- Y. Gichon[†], J. T. Jose[†], **H. Chandravamsi**, Y. Evron, & O. Ram (2024). *The dynamics of shock wave propagation far downstream of an abrupt area expansion*. Journal of Fluid Mechanics, 997, A30.
- **H. Chandravamsi**, D. V. Shenoy, I. Zinn, Z. Chen, S. Pisnoy, & S. H. Frankel (2025). *Spectral bottleneck in sinusoidal representation networks: noise is all you need*. Preprint under review. [PDF]
- Z. Chen, G. G. Shaviner, **H. Chandravamsi**, S. Pisnoy, S. H. Frankel, & U. Pereg (2026). *Quantum physics-informed neural networks for Maxwell's equations: circuit design, barren plateaus mitigation, and GPU acceleration*. Quantum Machine Intelligence, 8(1). [PDF]
- G. G. Shaviner, **H. Chandravamsi**, S. Pisnoy, Z. Chen, S. H. Frankel (2025). *PINNs for Solving Unsteady Maxwell's Equations: Convergence Issues and Comparative Assessment with Compact Schemes*. Neural Computing and Applications, 37(29) [PDF]
- J. F. Gueiros, **H. Chandravamsi**, & S. H. Frankel (2025). *Deep Learning vs. Black-Scholes: Option Pricing Performance on Brazilian Petrobras Stocks*. arXiv preprint. [PDF]
- **H. Chandravamsi**, A. S. Chamarthi, N. Hoffmann, S. H. Frankel (2023). *On the application of gradient based reconstruction for flow simulations on generalized curvilinear and dynamic mesh domains*. Computers & Fluids, 258, 105859. [DOI] [PDF]
- A. S. Chamarthi, **H. Chandravamsi**, N. Hoffmann, S. Bokor, S. H. Frankel (2023). *On the role of spectral properties of viscous flux discretization for flow simulations on marginally resolved grids*. Computers & Fluids, 251, 105742. [PDF]
- **H. Chandravamsi**, S. Bhardwaj, R. Sriram, K. Ramachandra (2023). *Control of bow shock induced three-dimensional separation using bleed through holes*. Physics of Fluids, 35, 016121. [PDF]
- S. Bhardwaj, **H. Chandravamsi**, R. Sriram (2022). *On the scaling of three-dimensional shock induced separated flow due to protuberances*. Physics of Fluids, 34, 076115. [PDF]
- **H. Chandravamsi**, N. R. Vadlamani, P. G. Tucker (2022). *On the use of high order central difference schemes for differential equation based wall distance computations*. Computers & Fluids, 248, 105666. [PDF]
- **H. Chandravamsi**, S. Pisnoy, D. V. Shenoy, S. H. Frankel. *Noisy Sine Layers To Mitigate Spectral Bias in Deep Networks*. (under preparation)

Conference Presentations, Proceedings, & Posters

- N. Hoffmann, A. S. Chamarthi, S. Bokor, A. Davidson, **H. Chandravamsi**, & S. H. Frankel (Jan 2026). *Wall-modeled large-eddy simulation of hypersonic boundary layer transition over a cone-cylinder-flare*, AIAA SciTech Forum 2026, AIAA 2026-0022.
- N. Hoffmann, A. S. Chamarthi, S. Bokor, A. Davidson, **H. Chandravamsi**, & S. H. Frankel (Jan 2026). *Effect of wall temperature on hypersonic boundary layer transition over a compression ramp*, AIAA SciTech Forum 2026, AIAA 2026-0023.
- Y. Gichon, J. T. Jose, R. Heppner, **H. Chandravamsi**, and O. Ram, *Shock wave interaction with an abrupt area expansion*, 1st Israel Fluid Mechanics Conference, 2025, Haifa, Israel. (Poster)
- R. Heppner, Y. Gichon, **H. Chandravamsi**, S. H. Frankel, and O. Ram, *Shock propagation through a localized narrowing*, 1st Israel Fluid Mechanics Conference, 2025, Haifa, Israel. (Poster)
- Y. Gichon, R. Heppner, **H. Chandravamsi**, and O. Ram, *Investigating the Dynamics of Transmitted and Reflected Shock Waves in Converging Ducts with Varying Contraction Geometries*, 35th International Symposium on Shock Waves 2025, Queensland, Australia.
- R. Heppner, Y. Gichon, **H. Chandravamsi**, S. H. Frankel, and O. Ram, *Shock propagation in a conduit with contoured walls*, 70th Annual Meeting of the Israel Physical Society, 2024, Haifa, Israel.
- **H. Chandravamsi** & S. H. Frankel (Jan 2025). *Dispersion Relation Preserving Optimized High-Order Straight and Mixed Second Derivative Schemes: Application to Fluid Flow Equations*, AIAA SciTech Forum 2025, Orlando, Florida, USA.
- **H. Chandravamsi** & S. H. Frankel (Oct 2023). *On the use of adaptive filtering for stability enhancement and shock capturing in flux reconstruction for compressible Euler flows*, 14th Asian Computational Fluid Dynamics Conference.
- Y. Gichon, J. T. Jose, **H. Chandravamsi**, Y. Evron, and O. Ram (Jul 2024). *On the Shock Wave Propagation Dynamics Following Sudden Area Expansion*, 25th International Shock Interaction Symposium 2024, Delft, The Netherlands.
- K. Ramachandra, **H. Chandravamsi**, S. Bhardwaj, & R. Sriram (Oct 2022). *Study of unsteadiness due to 3-D shock-boundary layer interaction in flow over a square-faced protuberance*, 24th International Shock Interaction Symposium, IIT Madras. [Video presentation]
- **H. Chandravamsi**, A. S. Chamarthi, & S. H. Frankel (Jan 2024). *Impinging wall curvature effects on flow and screech tones in choked under-expanded supersonic jets*, AIAA SciTech Forum 2024, Orlando, Florida, USA.
- **H. Chandravamsi**, D. K. Eswaran, & S. H. Frankel (Nov 2023). *Large eddy simulations of impinging supersonic jets: A close look at the unsteadiness*, V0042 - APS Gallery of Fluid Motion, 76th APS DFD. [Video]
- **H. Chandravamsi**, A. S. Chamarthi, & S. H. Frankel (Jul 2023). *Unsteady oscillation modes of choked under-expanded supersonic jets impinging on curved surfaces*, CFDIMPACT, Haifa, Israel.
- **H. Chandravamsi**, A. S. Chamarthi, & S. H. Frankel (Apr 2023). *High-resolution GPU-accelerated numerical simulations of supersonic impinging jet noise using gradient-based reconstruction*, IACM 22nd Computational Fluids Conference, Cannes, France.
- **H. Chandravamsi**, A. S. Chamarthi, & S. H. Frankel (Mar 2023). *Dynamical effects of impinging wall curvature on screech in under-expanded supersonic jets*, 62nd Israel Annual Conference on Aerospace Sciences. [Video presentation]
- **H. Chandravamsi**, A. S. Chamarthi, N. Hoffmann, & S. H. Frankel (Jan 2023). *GPU-accelerated numerical study of temperature effects in choked under-expanded supersonic jets*, AIAA SciTech Forum. [DOI]
- **H. Chandravamsi** & S. H. Frankel (Dec 2022). *Screech! These supersonic jets are hot*, Poster at Faculty of Mechanical Engineering - Research Day, Technion. [Poster]
- N. Hoffmann, A. S. Chamarthi, S. Bokor, **H. Chandravamsi**, & S. H. Frankel (Jul 2023). *GPU accelerated large eddy simulations of hypersonic boundary layer transition*, CFDIMPACT, Haifa, Israel.
- N. Hoffmann, A. S. Chamarthi, **H. Chandravamsi**, & S. H. Frankel (Jan 2023). *Large eddy simulation of transitional and turbulent hypersonic flow*, AIAA SciTech Forum. [DOI and video presentation]
- **H. Chandravamsi**, S. Bhardwaj, & R. Sriram (Jan 2020). *Control of protrusion induced unsteady supersonic separation using boundary layer suction*, PN-69, 6th National Symposium on Shock Waves.
- **H. Chandravamsi** & N. R. Vadlamani (Dec 2020). *A higher order differential equation based unsteady wall distance solver*, 8th International and 47th National Conference on Fluid Mechanics and Fluid Power. [Video presentation]

Open source repositories

1. PINNACLE: Open-source Repository For Advanced Multi-GPU Physics-Informed Machine Learning · Developers: H. Chandravamsi, Shimon Pisnoy, Aaron Goldgewert, Boris Shragner, Steven H. Frankel.
2. CAF-FR: GPU accelerated Cartesian flux reconstruction framework for high-speed shock containing flow simulations · Developers: H. Chandravamsi, Steven H. Frankel.

Invited Talks

1. *Numerical treatment of diffusion terms in compressible Navier-Stokes equations* · Guest lecture - Faculty of Mechanical Engineering, Technion (Feb 2025).
2. *High-performance parallelization strategies for computational fluid dynamics using OpenACC+MPI* · Guest lecture - Faculty of Mechanical Engineering, Technion (Dec 2022).
3. *GPU-accelerated CFD using OpenACC and application to supersonic jet noise prediction* · Rafael Advanced Defense Systems - Israel (Aug 2022).

Teaching Experience

1. **Teaching Assistant:** *Machine Learning for Scientists and Engineers* Winter 2023-2024 & 2024-2025
Instructor: Prof. Steven H. Frankel, Technion
Delivered coding tutorials on PyTorch and Physics-Informed Neural Networks (PINNs), created a github repository PINNACLE, a hands-on tool for new students of PINNs, provided guidance on projects and homework, contributed in assignment preparation, and grading.
2. **Laboratory Assistant:** *Gas Dynamics Laboratory* Spring 2019
Instructor: R. Sriram, IIT Madras
Set up weekly laboratory experiments and delivered student demonstrations involving schlieren imaging, shock tube experiments, supersonic high-speed wind experiments, and pitot tube measurements.
3. **Teaching Assistant:** *Compressible Fluid Flows* Winter 2019–2020
Instructor: R. Sriram, IIT Madras

Academic service and activities

1. **Peer reviewer:** Journal of Fluid Mechanics, Journal of Sound and Vibration, Aerospace Science and Technology, Journal of Open Source Software (JOSS), Physics of Fluids, Journal of Applied Physics, AIP Advances, and IEEE Access.
2. **Hackathon:** Team MIGHTY at OpenACC GPU Hackathon 2021, hosted by NVIDIA and Georgia Tech, USA.

Skills

1. **Programming:** Fortran, Python, MPI, OpenACC, OpenMP, Mathematica, MATLAB, bash, julia.
2. **Frameworks:** PyTorch, NumPy, Tensorflow.
3. **Experiments:** Oil-flow visualization, schlieren/shadowgraph, supersonic blowdown wind-tunnels, shock tubes.
4. **Analysis:** Fourier analysis, Von Neumann analysis, Proper Orthogonal Decomposition, Spectral Proper Orthogonal Decomposition, Dynamic Mode Decomposition, Correlation analysis.
5. **Software:** git, MacOS/Linux, Paraview (advanced post-processing), and Pointwise.

References

- **Tamer A. Zaki** (Postdoctoral advisor)
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